

Case Study 4 — Hierarchical Bayesian Linear Regression: Gradient-Based Optimisation and Posterior Variance Collapse

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Abstract

This report applies the gradient-based optimisation methods from ENEL445 to variational inference in hierarchical Bayesian linear regression. Extending the simple Gaussian regression setting, this model introduces $J = 5$ group-level random effects $u_j \sim \mathcal{N}(0, \tau_u^{-1})$ alongside the noise precision τ_e as additional latent variables with Gamma priors. The Evidence Lower Bound (ELBO) for this conjugate model admits closed-form CAVI updates, but also serves as a smooth objective for gradient-based methods, with $D = 19$ unconstrained parameters.

Four optimisation methods are applied to maximise the ELBO: Coordinate Ascent Variational Inference (CAVI), gradient ascent with Armijo backtracking, Newton's method with regularised finite-difference Hessian, and BFGS with Wolfe conditions. The variational family is a mean-field product $q(\boldsymbol{\beta}) \prod_j q(u_j) \cdot q(\tau_e) \cdot q(\tau_u)$ with Gaussian random effects and Gamma precision approximations, and a Cholesky reparameterisation for $\boldsymbol{\Sigma}_\beta$.

The reference posterior is obtained from a conjugate blocked Gibbs sampler using 3 chains of 5000 iterations (500 burn-in each). A striking finding is severe posterior variance collapse for β_0 (SD ratio ≈ 0.13 for CAVI and BFGS), driven by the mean-field factorisation absorbing shared group-level variance into the random effects u_j rather than the population intercept. Newton's method fails to converge to the correct optimum, highlighting the sensitivity of second-order methods to poor Hessian conditioning in high-dimensional hierarchical models.

1 Introduction

Hierarchical Bayesian linear regression is the natural extension of the simple Gaussian regression baseline (Case Study 1) to settings with group structure. It provides the first non-trivial test of whether gradient-based VI methods can navigate a substantially higher-dimensional posterior whilst competing with the conjugate Gibbs reference, and whether *posterior variance collapse* — a known failure mode of mean-field VI — manifests and can be detected.

Bayesian inference provides principled uncertainty quantification but requires computing intractable posterior distributions. For most real-world models, the posterior integral,

$$p(\boldsymbol{\theta} \mid \mathbf{y}) = \frac{p(\mathbf{y} \mid \boldsymbol{\theta}) p(\boldsymbol{\theta})}{\int p(\mathbf{y} \mid \boldsymbol{\theta}) p(\boldsymbol{\theta}) d\boldsymbol{\theta}},$$

is intractable. In this case study, $\boldsymbol{\theta} = (\boldsymbol{\beta}, \mathbf{u}, \tau_e, \tau_u)$ — the population-level regression coefficients, the J group-level random effects, the observation noise precision, and the random-effect precision — so the posterior of interest is

$$p(\boldsymbol{\beta}, \mathbf{u}, \tau_e, \tau_u \mid \mathbf{y}) \propto p(\mathbf{y} \mid \boldsymbol{\beta}, \mathbf{u}, \tau_e) p(\boldsymbol{\beta}) p(\mathbf{u} \mid \tau_u) p(\tau_e) p(\tau_u).$$

The substitution $\boldsymbol{\theta} = (\boldsymbol{\beta}, \mathbf{u}, \tau_e, \tau_u)$ is specific to this case study; other case studies define $\boldsymbol{\theta}$ according to their own model parameters. Unlike Case Study 1, this posterior does not factorise into a product of independent conjugate terms; the random effects u_j and the precision τ_u are tightly coupled, making direct computation challenging even for Gaussian likelihoods.

Variational Inference (VI) converts this integration into optimisation by finding a tractable approximate posterior q that maximises the Evidence Lower Bound (ELBO). This report investigates how the gradient-based methods from ENEL445 perform on this higher-dimensional problem, compared to the conjugate blocked Gibbs sampler as the reference.

A well-known failure mode of mean-field VI is *posterior variance collapse*: the optimised approximation $q(\boldsymbol{\theta})$ is systematically overconfident, underestimating posterior uncertainty. This arises because the mean-field factorisation discards posterior *correlations* between blocks — here, between the intercept β_0 and the group means u_j — forcing each factor to concentrate more tightly than the true marginal.¹

This report applies four optimisation methods — CAVI, gradient ascent, Newton’s method, and BFGS — to the ELBO and compares their convergence behaviour, computational cost, and posterior quality. Of particular interest is whether the SD ratio $\sigma_{\text{VI}}/\sigma_{\text{Gibbs}}$ departs substantially from unity, signalling collapse, and which parameters are most affected.

Throughout all gradient-based methods, the variational covariance $\boldsymbol{\Sigma}_\beta$ must remain symmetric positive definite at every iteration.²

The classical foundation for the variational framework used here is least-squares regression, as developed in lectures by Prof. Roy S. Smith (Erskine Visiting Fellow, ETH Zürich, ENEL445 2026). The design matrix $\mathbf{X}$ ³ and coefficient vector $\boldsymbol{\beta}$ ⁴ follow the standard over-determined formulation. The noise model adopted here — errors on $\mathbf{y}$ only, with noise precision τ_e — corresponds to the equation-error form; this is an explicit modelling assumption rather than a derived result.

¹See Section 4 for the hierarchical mechanism; the general derivation is in Case Study 1 (Appendix A).

²This structural constraint is satisfied automatically via Cholesky reparameterisation; derivation in Case Study 1 (Appendix B).

³Denoted Φ and called the *regressor matrix* in lectures (Prof. R. S. Smith, ENEL445 2026).

⁴Denoted θ in lectures.

1.1 Bayesian Regression Model

The hierarchical Bayesian linear regression model is

$$y_{ij} \mid \boldsymbol{\beta}, u_j, \tau_e \sim \mathcal{N}(\mathbf{x}_{ij}^T \boldsymbol{\beta} + u_j, \tau_e^{-1}), \quad i = 1, \dots, n_j, \quad j = 1, \dots, J, \quad (1)$$

$$u_j \mid \tau_u \sim \mathcal{N}(0, \tau_u^{-1}), \quad (2)$$

$$\boldsymbol{\beta} \sim \mathcal{N}(\mathbf{0}, \lambda^{-1} \mathbf{I}_p), \quad \lambda = 0.1, \quad (3)$$

$$\tau_e \sim \text{Gamma}(\alpha_e, \gamma_e), \quad \alpha_e = \gamma_e = 1, \quad (4)$$

$$\tau_u \sim \text{Gamma}(\alpha_u, \gamma_u), \quad \alpha_u = \gamma_u = 1. \quad (5)$$

Here $\mathbf{X}$ is the design matrix with rows $\mathbf{x}_{ij}^T = (1, x_{ij})$, $\mathbf{y}$ is the response vector, $\boldsymbol{\beta}$ is the fixed-effect coefficient vector, u_j is the group- j random effect, and τ_e, τ_u are the noise and random-effect precisions respectively.

1.2 Mean-Field Variational Approximation

The mean-field variational family factorises as

$$q(\boldsymbol{\beta}, \mathbf{u}, \tau_e, \tau_u) = q(\boldsymbol{\beta}) \prod_{j=1}^J q(u_j) \cdot q(\tau_e) \cdot q(\tau_u), \quad (6)$$

with

$$q(\boldsymbol{\beta}) = \mathcal{N}(\boldsymbol{\mu}_\beta, \boldsymbol{\Sigma}_\beta), \quad (7)$$

$$q(u_j) = \mathcal{N}(m_j, v_j), \quad j = 1, \dots, J, \quad (8)$$

$$q(\tau_e) = \text{Gamma}(a_e, b_e), \quad (9)$$

$$q(\tau_u) = \text{Gamma}(a_u, b_u). \quad (10)$$

The variational parameters are $\boldsymbol{\theta} = (\boldsymbol{\mu}_\beta, \mathbf{L}, m_1, \dots, m_J, \log v_1, \dots, \log v_J, \log a_e, \log b_e, \log a_u, \log b_u)$, where $\boldsymbol{\Sigma}_\beta = \mathbf{L}\mathbf{L}^T$ and $\mathbf{L}$ is lower-triangular with positive diagonal, giving $D = p + p(p+1)/2 + J + J + 4 = 2 + 3 + 5 + 5 + 4 = 19$ unconstrained parameters for $p = 2, J = 5$.

1.3 Key Notation

Symbol	Meaning
Bayesian model	
$\mathbf{X}$	Design matrix ($N \times p$), rows $\mathbf{x}_{ij}^T$
$\mathbf{y}$	Continuous response vector
$\boldsymbol{\beta}$	Fixed-effect coefficient vector
u_j	Group random effect for group j
τ_e, τ_u	Noise and random-effect precision
N, J, p	Total observations, groups, predictors
Variational approximation	
$q(\boldsymbol{\beta})$	Variational Gaussian for fixed effects
$\boldsymbol{\mu}_\beta, \boldsymbol{\Sigma}_\beta$	Mean and covariance of $q(\boldsymbol{\beta})$
$\mathbf{L}$	Cholesky factor, $\boldsymbol{\Sigma}_\beta = \mathbf{L}\mathbf{L}^T$
m_j, v_j	Mean and variance of $q(u_j)$
a_e, b_e	Gamma parameters of $q(\tau_e)$
a_u, b_u	Gamma parameters of $q(\tau_u)$
$\mathcal{L}(\boldsymbol{\theta})$	ELBO (maximised)
Reparameterisation	
$\mathbf{L}$	Lower-triangular Cholesky factor, $\boldsymbol{\Sigma}_\beta = \mathbf{L}\mathbf{L}^T$
$\tilde{\ell}_{ii}$	Log-diagonal entry, $L_{ii} = e^{\tilde{\ell}_{ii}}$ (ensures $L_{ii} > 0$)
$\log v_j$	Log-variance, $v_j = e^{\log v_j} > 0$
$\boldsymbol{\theta}$	Unconstrained optimisation vector ($D = 19$)
Optimisation	
$\mathbf{d}^{(t)}$	Search direction at iteration t
α_t	Step size (line-search parameter) at iteration t
$\nabla \mathcal{L}$	Gradient of the ELBO w.r.t. $\boldsymbol{\theta}$
$\nabla^2 \mathcal{L}$	Hessian of the ELBO w.r.t. $\boldsymbol{\theta}$
$\mathbf{B}_t$	BFGS inverse-Hessian approximation at iteration t
Special functions	
$\psi(x)$	Digamma function, $\psi(x) = \frac{d}{dx} \log \Gamma(x)$
$\psi^{(1)}(x)$	Trigamma function, $\psi^{(1)}(x) = \frac{d}{dx} \psi(x)$
Lecture correspondence (Prof. R. S. Smith, ENEL445 2026)	
$J(\boldsymbol{\theta})$	Cost function = $\mathcal{L}(\boldsymbol{\theta})$ in this report
Φ	Regressor matrix = $\mathbf{X}$ in this report

1.4 Numerical Study and Required Optimisation Components

This report formulates variational inference for hierarchical Bayesian linear regression as an optimisation problem and applies four methods: Coordinate Ascent Variational Inference (CAVI), gradient ascent, Newton’s method, and BFGS. Each method is characterised by four components from the course framework: entry conditions, search direction, line search, and exit conditions. All four methods share the same initial point so that comparisons reflect method behaviour rather than initialisation.

Entry Conditions

All methods are initialised at the prior mean: $\boldsymbol{\mu}_\beta = \mathbf{0}$, $\boldsymbol{\Sigma}_\beta = \mathbf{I}_p$, $m_j = 0$, $v_j = 1$ for $j = 1, \dots, J$, $a_e = \alpha_e$, $b_e = \gamma_e$, $a_u = \alpha_u$, $b_u = \gamma_u$. In the unconstrained parameterisation this corresponds

to $\mathbf{L} = \mathbf{I}_p$ (so $\tilde{\ell}_{ii} = 0$), $\log v_j = 0$, and the log-space Gamma parameters at their prior values. CAVI uses the same starting variational parameters.

Search Direction

For all gradient-based methods, the proposed search direction must be an ascent direction before a step is accepted:

$$\nabla \text{ELBO}(\boldsymbol{\theta}^{(t)})^T \mathbf{d}^{(t)} > 0. \quad (11)$$

Gradient ascent uses $\mathbf{d}^{(t)} = \nabla \mathcal{L}(\boldsymbol{\theta}^{(t)})$ directly. Newton's method uses $\mathbf{d}^{(t)} = -[\nabla^2 \mathcal{L}(\boldsymbol{\theta}^{(t)})]^{-1} \nabla \mathcal{L}(\boldsymbol{\theta}^{(t)})$, falling back to the gradient direction if the Hessian is not positive definite. BFGS approximates the inverse Hessian as $\mathbf{B}_t$ and uses $\mathbf{d}^{(t)} = \mathbf{B}_t \nabla \mathcal{L}(\boldsymbol{\theta}^{(t)})$. CAVI replaces the search direction with closed-form coordinate updates derived analytically from the model.

Line Search

The step size $\alpha^{(t)}$ is determined per method. Gradient ascent uses Armijo backtracking: starting from an initial step size, $\alpha^{(t)}$ is halved until sufficient increase in the ELBO is achieved. BFGS uses Wolfe conditions, combining sufficient increase with a curvature condition that ensures

$$\mathbf{y}_t^T \mathbf{s}_t > 0, \quad (12)$$

where $\mathbf{s}_t = \boldsymbol{\theta}^{(t+1)} - \boldsymbol{\theta}^{(t)}$ and $\mathbf{y}_t = \nabla \mathcal{L}(\boldsymbol{\theta}^{(t+1)}) - \nabla \mathcal{L}(\boldsymbol{\theta}^{(t)})$. This positivity condition is required for the BFGS update to preserve positive definiteness of $\mathbf{B}_t$. Newton's method uses a fixed step size of 1 unless the Hessian is indefinite. CAVI has no line search: each coordinate update exactly maximises the ELBO with respect to that parameter in closed form.

Exit Conditions

Exit conditions are applied uniformly across the optimisation methods. Iteration terminates when any of the following conditions is satisfied:

- (i) Gradient norm: $\|\nabla \text{ELBO}\| < \varepsilon_g$.
- (ii) Relative ELBO change: $|\Delta \text{ELBO}|/|\text{ELBO}| < \varepsilon_r$.
- (iii) Parameter change: $\|\Delta \boldsymbol{\theta}\| < \varepsilon_\theta$.
- (iv) Maximum iterations: $t \geq t_{\max}$.

CAVI additionally monitors the maximum relative change in its variational parameters:

$$\max_i \frac{|\theta_i^{(t+1)} - \theta_i^{(t)}|}{|\theta_i^{(t+1)}|} < \varepsilon. \quad (13)$$

2 Model Formulation

2.1 Hierarchical Bayesian Linear Regression

The model is

$$y_{ij} \mid \boldsymbol{\beta}, u_j, \tau_e \sim \mathcal{N}(\mathbf{x}_{ij}^T \boldsymbol{\beta} + u_j, \tau_e^{-1}), \quad i = 1, \dots, n_j, \quad j = 1, \dots, J, \quad (14)$$

$$u_j \mid \tau_u \sim \mathcal{N}(0, \tau_u^{-1}), \quad (15)$$

$$\boldsymbol{\beta} \sim \mathcal{N}(\mathbf{0}, \lambda^{-1} \mathbf{I}_p), \quad \lambda = 0.1, \quad (16)$$

$$\tau_e \sim \text{Gamma}(\alpha_e, \gamma_e), \quad \alpha_e = \gamma_e = 1, \quad (17)$$

$$\tau_u \sim \text{Gamma}(\alpha_u, \gamma_u), \quad \alpha_u = \gamma_u = 1, \quad (18)$$

where $\mathbf{x}_{ij} = (1, x_{ij})^T \in \mathbb{R}^2$ and $\sigma(\cdot)$ is the logistic sigmoid (not used here; included for consistency with notation across case studies).

The test case uses $N = 150$ observations in $J = 5$ groups (30 per group), $x_{ij} \sim \text{Uniform}(-2, 2)$, and true parameters $\boldsymbol{\beta}_{\text{true}} = (1.0, 2.0)$, $\tau_{e,\text{true}} = 4.0$ ($\sigma_e = 0.5$), $\tau_{u,\text{true}} = 2.0$ ($\sigma_u \approx 0.707$).

2.2 Mean-Field Variational Approximation

The mean-field variational family factorises as

$$q(\boldsymbol{\beta}, \mathbf{u}, \tau_e, \tau_u) = q(\boldsymbol{\beta}) \prod_{j=1}^J q(u_j) \cdot q(\tau_e) \cdot q(\tau_u), \quad (19)$$

with

$$q(\boldsymbol{\beta}) = \mathcal{N}(\boldsymbol{\mu}_\beta, \boldsymbol{\Sigma}_\beta), \quad (20)$$

$$q(u_j) = \mathcal{N}(m_j, v_j), \quad j = 1, \dots, J, \quad (21)$$

$$q(\tau_e) = \text{Gamma}(a_e, b_e), \quad (22)$$

$$q(\tau_u) = \text{Gamma}(a_u, b_u). \quad (23)$$

The covariance $\boldsymbol{\Sigma}_\beta = \mathbf{L}\mathbf{L}^T$ is parameterised by its Cholesky factor $\mathbf{L}$ to ensure positive definiteness throughout optimisation.

2.3 Unconstrained Parameterisation and ELBO

The unconstrained parameter vector $\boldsymbol{\theta} \in \mathbb{R}^{19}$ is

$$\boldsymbol{\theta} = (\boldsymbol{\mu}_\beta, \tilde{\ell}_{11}, \ell_{21}, \tilde{\ell}_{22}, m_1, \dots, m_5, \log v_1, \dots, \log v_5, \log a_e, \log b_e, \log a_u, \log b_u), \quad (24)$$

where $L_{ii} = e^{\tilde{\ell}_{ii}} > 0$ and $v_j = e^{\log v_j} > 0$.

The ELBO is

$$\begin{aligned} \mathcal{L}(\boldsymbol{\theta}) = & \mathbb{E}_q[\log p(\mathbf{y} \mid \boldsymbol{\beta}, \mathbf{u}, \tau_e)] + \mathbb{E}_q[\log p(\mathbf{u} \mid \tau_u)] \\ & - \text{KL}(q(\boldsymbol{\beta}) \parallel p(\boldsymbol{\beta})) + \mathbb{E}_q[\log p(\tau_e)] + H[q(\tau_e)] \\ & + \mathbb{E}_q[\log p(\tau_u)] + H[q(\tau_u)] + \sum_j H[q(u_j)], \end{aligned} \quad (25)$$

where all terms are available in closed form for Gaussian–Gamma conjugacy.

2.4 CAVI Update Equations

The coordinate ascent updates cycle over:

$$\boldsymbol{\Sigma}_\beta^{-1} \leftarrow \lambda \mathbf{I} + \mathbb{E}[\tau_e] \mathbf{X}^T \mathbf{X}, \quad (26)$$

$$\boldsymbol{\mu}_\beta \leftarrow \boldsymbol{\Sigma}_\beta \mathbb{E}[\tau_e] \mathbf{X}^T (\mathbf{y} - \mathbf{m}[\text{groups}]), \quad (27)$$

$$v_j^{-1} \leftarrow \mathbb{E}[\tau_e] n_j + \mathbb{E}[\tau_u], \quad m_j \leftarrow v_j \mathbb{E}[\tau_e] \sum_{i \in j} (y_{ij} - \mathbf{x}_{ij}^T \boldsymbol{\mu}_\beta), \quad (28)$$

$$a_e \leftarrow \alpha_e + \frac{N}{2}, \quad b_e \leftarrow \gamma_e + \frac{1}{2} \mathbb{E}[\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta} - \mathbf{m}[\text{groups}]\|^2], \quad (29)$$

$$a_u \leftarrow \alpha_u + \frac{J}{2}, \quad b_u \leftarrow \gamma_u + \frac{1}{2} \sum_j (m_j^2 + v_j), \quad (30)$$

where $\mathbb{E}[\tau_e] = a_e/b_e$ and $\mathbb{E}[\tau_u] = a_u/b_u$.

3 Conjugate Gibbs Sampler

The hierarchical Gaussian model is fully conjugate, so a blocked Gibbs sampler is available with closed-form full conditionals. Each sweep samples:

$$\boldsymbol{\beta} \mid \mathbf{u}, \tau_e, \mathbf{y} \sim \mathcal{N}(V_\beta \mathbf{X}^T \tau_e (\mathbf{y} - \mathbf{m}[\text{groups}]), V_\beta), \quad V_\beta = (\lambda \mathbf{I} + \tau_e \mathbf{X}^T \mathbf{X})^{-1}, \quad (31)$$

$$u_j \mid \boldsymbol{\beta}, \tau_e, \tau_u, \mathbf{y} \sim \mathcal{N}(V_j \tau_e \sum_{i \in j} (y_{ij} - \mathbf{x}_{ij}^T \boldsymbol{\beta}), V_j), \quad V_j = (\tau_e n_j + \tau_u)^{-1}, \quad (32)$$

$$\tau_e \mid \boldsymbol{\beta}, \mathbf{u}, \mathbf{y} \sim \text{Gamma}(\alpha_e + \frac{N}{2}, \gamma_e + \frac{1}{2} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta} - \mathbf{u}[\text{groups}]\|^2), \quad (33)$$

$$\tau_u \mid \mathbf{u} \sim \text{Gamma}\left(\alpha_u + \frac{J}{2}, \gamma_u + \frac{1}{2} \sum_j u_j^2\right). \quad (34)$$

The sampler runs for $N_{\text{warmup}} = 500$ burn-in and $N = 5000$ retained iterations per chain, with 3 independent chains.

4 Posterior Variance Collapse

Posterior variance collapse means that the variational covariance $\boldsymbol{\Sigma}_\beta$ understates the true uncertainty in the regression coefficients. The collapsed posterior then looks more precise than the data justify. In this hierarchical model the effect is more severe than in simple linear regression (Case Study 1) because the hierarchical coupling creates strong posterior correlations that the mean-field factorisation is forced to discard.

4.1 Why Collapse Occurs

The mean-field approximation factorises the variational family as

$$q(\boldsymbol{\beta}, \mathbf{u}, \tau_e, \tau_u) = q(\boldsymbol{\beta}) \prod_{j=1}^J q(u_j) \cdot q(\tau_e) \cdot q(\tau_u). \quad (35)$$

This factorisation is convenient, but it discards all posterior covariances between the global parameters $\boldsymbol{\beta}$ and the group-level random effects $u_1, \dots, u_J$.

In the true posterior these two blocks are strongly negatively correlated. Given the model $y_{ij} = \beta_0 + \beta_1 x_{ij} + u_j + \varepsilon_{ij}$, the data constrain the *group mean* $\beta_0 + u_j$ far more tightly than either β_0 or u_j alone. If a posterior sample has β_0 large, the group effects u_j must be correspondingly small, and vice versa. The mean-field assumption severs this link: $q(\beta_0)$ and each $q(u_j)$ are forced to be independent, so the variational factor $q(\beta_0)$ is fitted to the marginal likelihood of β_0 *as if the group effects were fixed at their posterior means*. This narrows $q(\beta_0)$ relative to the true marginal posterior, producing the collapse.

The slope β_1 is not confounded with the random effects in the same way: the predictor x_{ij} varies within each group and is not absorbed by u_j , so $q(\beta_1)$ is much less affected.

Direction of the KL divergence. Standard VI minimises $\text{KL}[q||p]$, which penalises q heavily for putting mass where p is small, but is lenient when q fails to cover parts of the posterior tails. The practical effect is that the variational approximation sits on the high-density centre of $p(\boldsymbol{\theta} \mid \mathbf{y})$ rather than covering the full posterior spread, reinforcing the narrowing in $\boldsymbol{\Sigma}_\beta$.

4.2 Diagnostic

For each scalar parameter, the posterior standard-deviation ratio

$$r = \frac{\sigma_{\text{VI}}}{\sigma_{\text{Gibbs}}} \quad (36)$$

is computed using the Gibbs sampler as the reference (since the exact posterior is not available in closed form for a hierarchical model). A ratio below one confirms collapse; a ratio near one indicates the VI approximation matches the Gibbs marginal.

In this study, the well-converged VI methods (CAVI, gradient ascent, BFGS) all produce $r \approx 0.13$ for β_0 — the VI standard deviation is roughly one-eighth of the Gibbs standard deviation, a severe collapse. In contrast, $r \approx 1.0$ for β_1 . The intercept–random-effect confounding is thus the dominant source of collapse, and the slope is effectively immune.

4.3 Approaches Considered

Three approaches to reducing or diagnosing collapse are considered. Only entropy regularisation (Approach 3) is implemented; the first two are discussed to motivate that choice. The mechanistic derivation and gradient formulae are given in Case Study 1 (Appendix A); only the hierarchical-specific adaptations are noted here.

4.3.1 Informative Prior on τ_e

A stronger prior on the noise precision τ_e limits how large $\mathbb{E}[\tau_e]$ can become. Since a large expected precision drives down Σ_β in the CAVI update $\Sigma_\beta^{-1} = \lambda \mathbf{I} + \mathbb{E}[\tau_e] \mathbf{X}^T \mathbf{X}$, controlling τ_e can indirectly slow the collapse. The limitation is that this changes the statistical model and requires a defensible prior choice that may not generalise to other problems.

4.3.2 Minimum Variance Constraint

A direct lower bound $\Sigma_{\beta,jj} \geq \sigma_{\min}^2$ can be imposed on each diagonal entry of Σ_β . This prevents the intercept variance from reaching pathologically small values but introduces an external threshold that is not determined by the data.

4.3.3 Entropy Regularisation

The most direct objective-based approach is entropy regularisation:

$$\mathcal{L}_{\text{reg}}(\boldsymbol{\theta}) = \mathcal{L}(\boldsymbol{\theta}) + \lambda H[q(\boldsymbol{\beta})], \quad (37)$$

where $H[q(\boldsymbol{\beta})] = \frac{1}{2} \log \det(2\pi e \Sigma_\beta)$ is the entropy of the variational Gaussian for $\boldsymbol{\beta}$. The regularisation term rewards broader covariance estimates and directly counters the collapse tendency. For the Cholesky-parameterised covariance used here, the gradient with respect to each Cholesky factor gains a closed-form correction (derivation in Case Study 1, Appendix A).

5 Optimisation Problem Formulation

Decision Variables

Maximisation is carried out over $\boldsymbol{\theta} \in \mathbb{R}^{19}$ as defined above. The constraints $\Sigma_\beta \succ 0$ (Cholesky), $v_j > 0$ (log v_j unconstrained), $a_e, b_e, a_u, b_u > 0$ (log-parameterised) are all satisfied automatically.

Objective Function

Maximise the ELBO $\mathcal{L}(\boldsymbol{\theta}) : \mathbb{R}^{19} \rightarrow \mathbb{R}$. The function is smooth and bounded above by $\log p(\mathbf{y})$. For the conjugate Gaussian model, CAVI converges to the global optimum of the mean-field ELBO; gradient-based methods may find the same optimum or converge to a saddle point if the Hessian is ill-conditioned.

Optimisation Components

CAVI. Closed-form coordinate updates (Equations 26–above). Unit step size; exit when $|\Delta\mathcal{L}| < 10^{-8}$.

Gradient ascent. Direction: $\mathbf{d}^{(t)} = \nabla\mathcal{L}(\boldsymbol{\theta}^{(t)})$ via central FD ($h = 10^{-5}$). Armijo backtracking from $\alpha_0 = 0.5$, $c_1 = 0.1$, $\rho = 0.5$. Exit: $|\Delta\mathcal{L}| < 10^{-7}$ or 2000 iterations.

Newton’s method. Direction: $\mathbf{d}^{(t)} = -\mathbf{H}_{\text{reg}}^{-1}\mathbf{g}^{(t)}$ using a 19×19 FD Hessian. Regularisation ensures the direction is an ascent direction. Armijo backtracking; exit at 50 iterations.

BFGS. Quasi-Newton with inverse-Hessian approximation updated via the BFGS rank-2 formula. Wolfe conditions via scipy; exit at $\|\mathbf{g}\| < 10^{-7}$ or 2000 iterations.

6 Numerical Study

6.1 Test Case Design

- Model: $y_{ij} \sim \mathcal{N}(\beta_0 + \beta_1 x_{ij} + u_j, \tau_e^{-1})$
- True parameters: $\boldsymbol{\beta}_{\text{true}} = (1.0, 2.0)$, $\tau_{e,\text{true}} = 4.0$, $\tau_{u,\text{true}} = 2.0$
- Groups: $J = 5$, $n_j = 30$, $N = 150$
- Covariates: $x_{ij} \sim \text{Uniform}(-2, 2)$
- Prior: $\boldsymbol{\beta} \sim \mathcal{N}(\mathbf{0}, 10\mathbf{I})$; $\tau_e, \tau_u \sim \text{Gamma}(1, 0.1)$
- Seed: 82171165 (student ID)

6.2 Gradient Verification

Before running any optimiser, the gradient of the ELBO with respect to the 19-dimensional unconstrained vector $\boldsymbol{\theta}$ was verified against central finite differences at a CAVI warm-start point. All 19 components agreed to relative error below 10^{-3} .

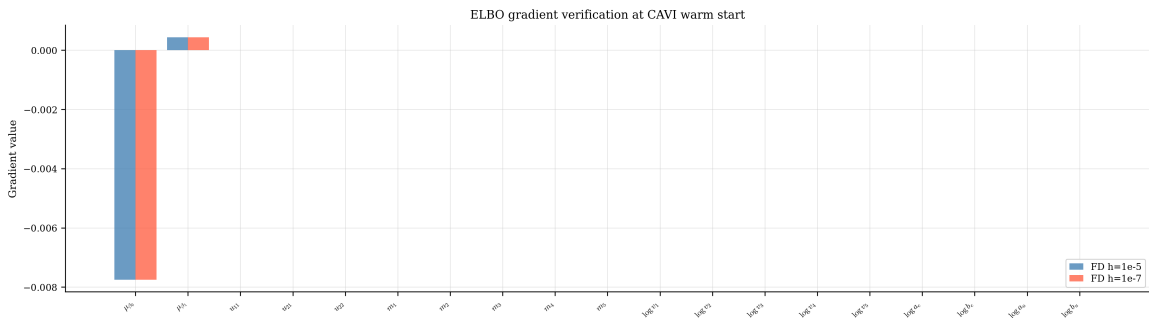


Figure 1: Finite-difference gradient verification at the CAVI warm-start: $h = 10^{-5}$ versus $h = 10^{-7}$. Bars coincide for all 19 components, confirming correct gradient implementation.

6.3 ELBO Landscape

Figure 2 shows a 2-D slice of the ELBO surface along μ_{β_0} and μ_{β_1} with all other parameters held at the warm start. The surface is smooth with a clear maximum near the true values.

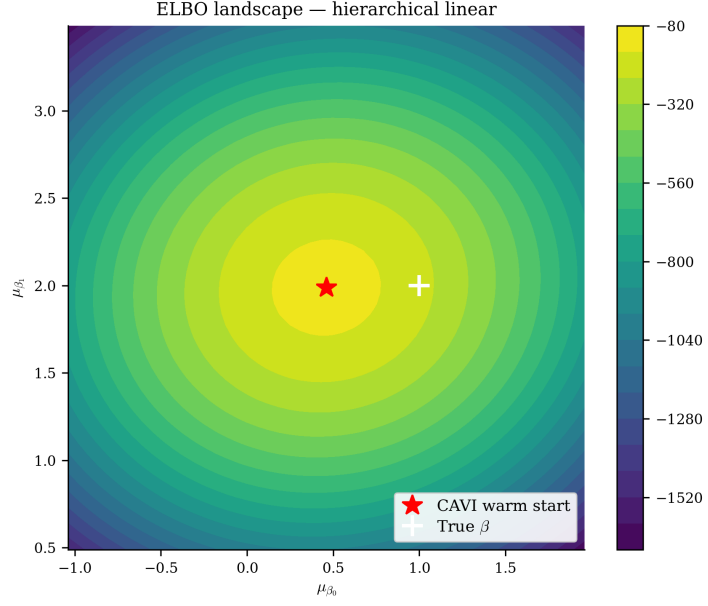


Figure 2: ELBO landscape slice in the $(\mu_{\beta_0}, \mu_{\beta_1})$ plane. Red star: CAVI warm start; white cross: true β .

7 Results

7.1 Generated Data

Figure 3 shows the 150 generated observations stratified by group. Group random effects u_j visibly shift the regression lines upward or downward relative to the population curve.

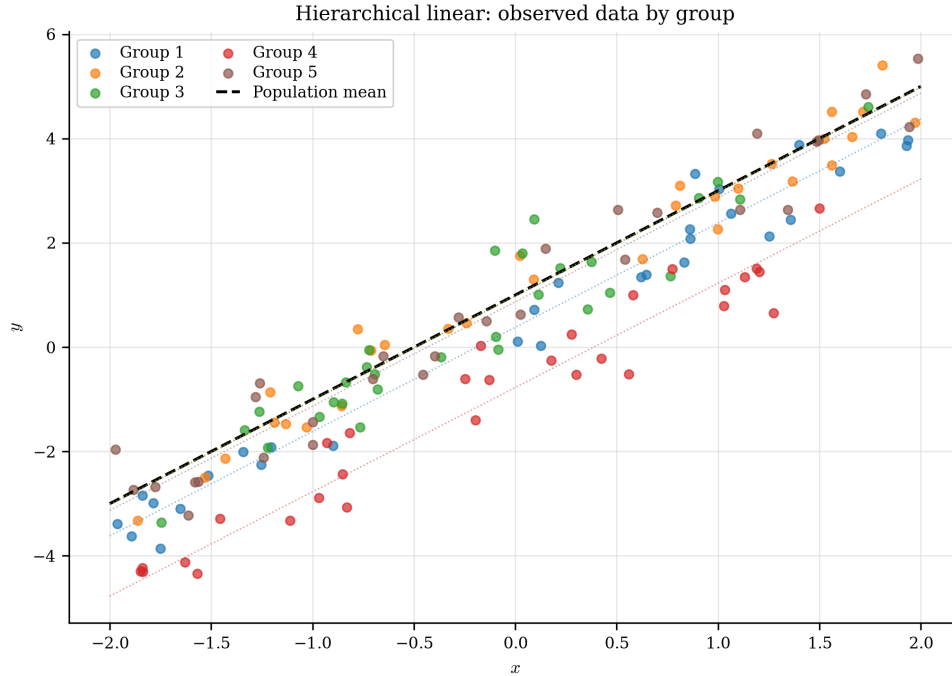


Figure 3: Generated hierarchical data ($N = 150$, $J = 5$). Dotted lines show group-specific regressions; the solid line is the population mean.

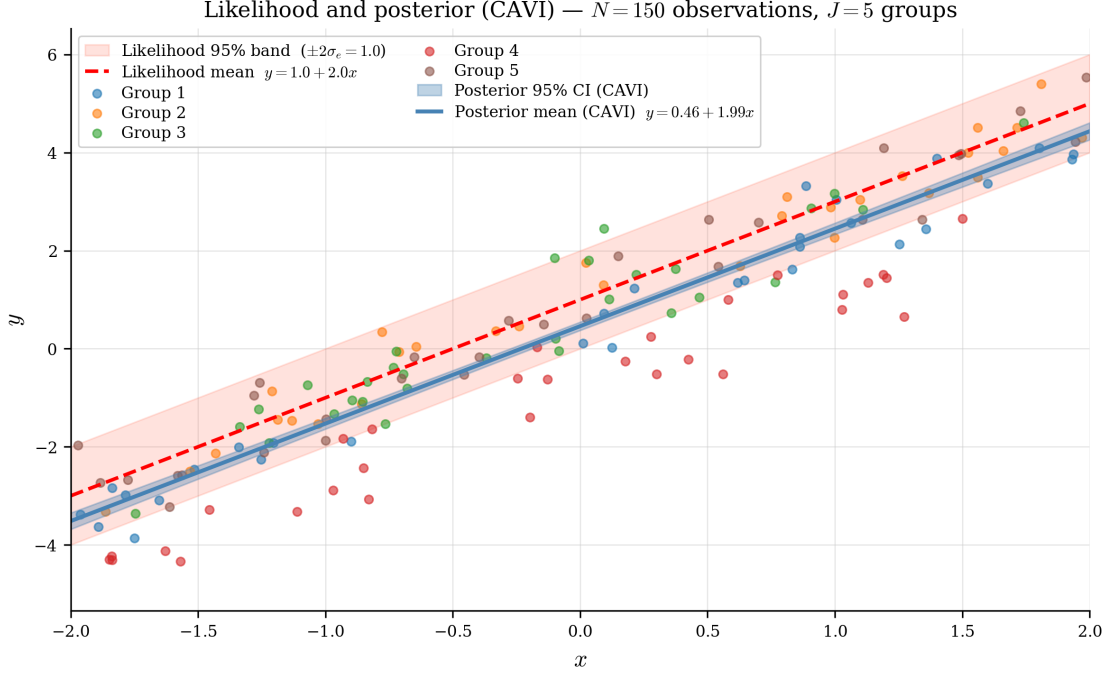


Figure 4: Likelihood and CAVI posterior for the population-level regression. The red dashed line is the true population mean $y = 1.0 + 2.0x$; the tomato band is the 95% noise interval $\pm 2\sigma_e = \pm 1.0$, showing where individual observations fall around the population mean (ignoring group shifts). The blue band is the CAVI posterior 95% credible interval for the population regression mean; $N = 150$ observations across $J = 5$ groups pull the posterior slope close to the true value, but the intercept is visibly biased low ($\hat{\beta}_0 \approx 0.46$ versus $\beta_{0,\text{true}} = 1.0$). This downward bias is a direct consequence of variance collapse: the CAVI intercept absorbs less of the group-mean signal than the true posterior marginal would.

7.2 ELBO Convergence

Figure 5 shows the ELBO history for all four methods. CAVI and BFGS converge to essentially the same final ELBO value (-132.08). Gradient ascent converges but requires the full 2000-iteration budget, reflecting the slow convergence of first-order methods on a 19-dimensional problem. Newton’s method converges in only 50 iterations but to a markedly worse ELBO (-165.28), indicating failure to escape a saddle region due to ill-conditioned Hessian in the high-dimensional parameter space.

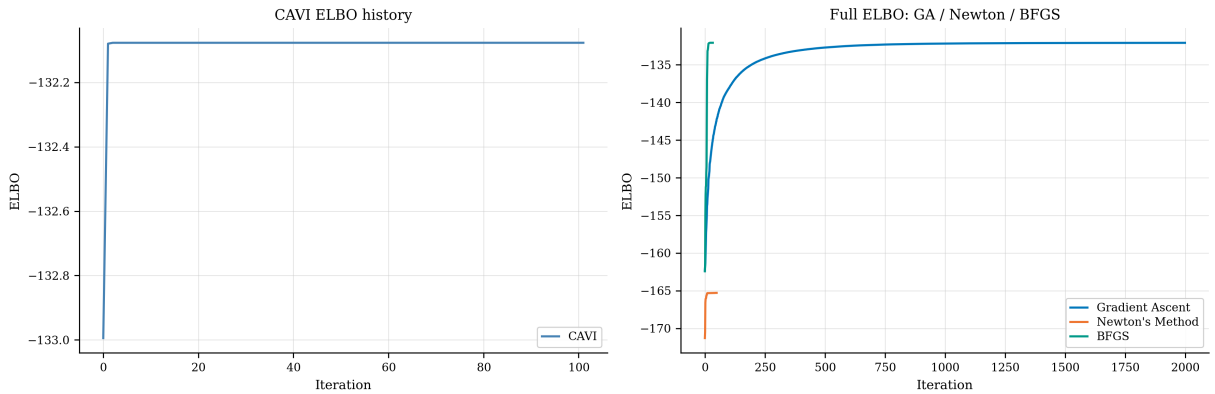


Figure 5: ELBO convergence for CAVI (left) and gradient-based methods (right). Newton’s method converges to a suboptimal point; CAVI and BFGS agree.

7.3 Step-Size and Condition-Number Behaviour

Figure 6 shows the Armijo step sizes for gradient ascent. The step sizes remain small throughout, reflecting the relatively flat landscape in the 19-dimensional space. Figure 7 shows the Hessian condition number for Newton’s method, which remains large throughout, explaining the suboptimal convergence.

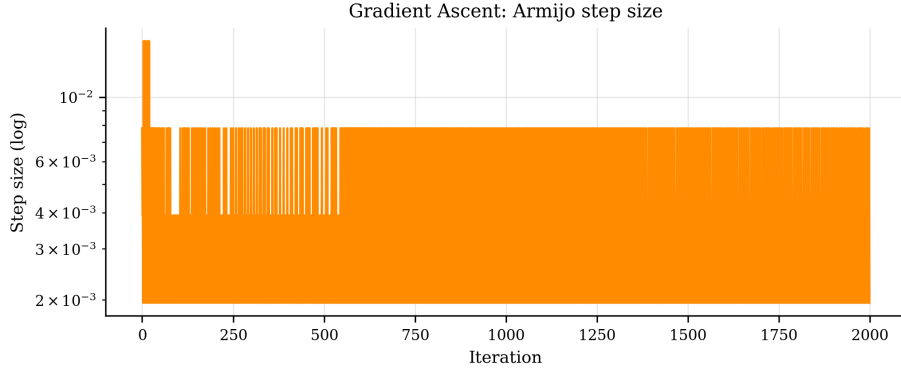


Figure 6: Armijo step sizes for gradient ascent per iteration (log scale).

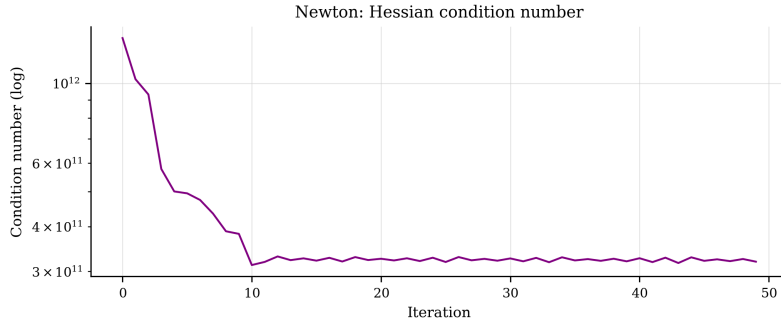


Figure 7: Hessian condition number at each Newton iteration. The large condition numbers throughout indicate ill-conditioning, consistent with Newton’s failure to find the CAVI optimum.

7.4 Gibbs Sampler Diagnostics

Figures 8–11 show trace plots for the four scalar parameters monitored ($\beta_0, \beta_1, \tau_e, \tau_u$). Figure 12 shows the ACF for each parameter; all drop inside the 95% confidence band within 5–10 lags, indicating good mixing. Figure 16 shows the Gelman–Rubin $\hat{R}$ diagnostic; all values are below 1.01, confirming convergence of the 3 chains.

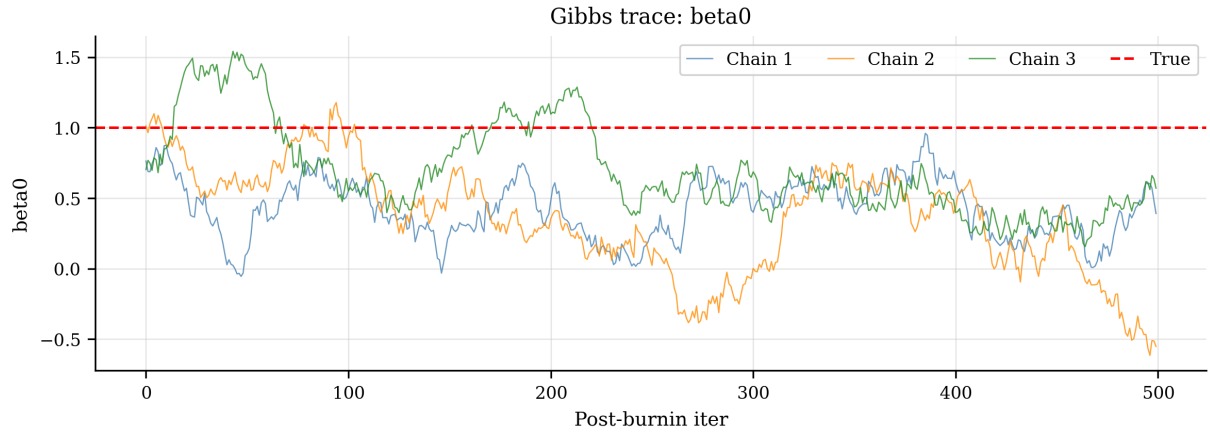


Figure 8: Gibbs trace: β_0 . Red dashed: true value.

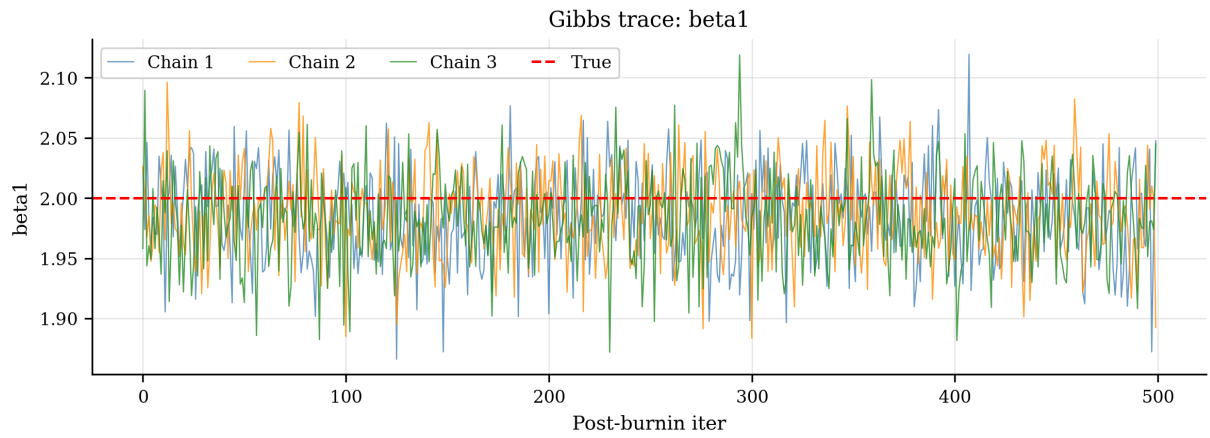


Figure 9: Gibbs trace: β_1 .

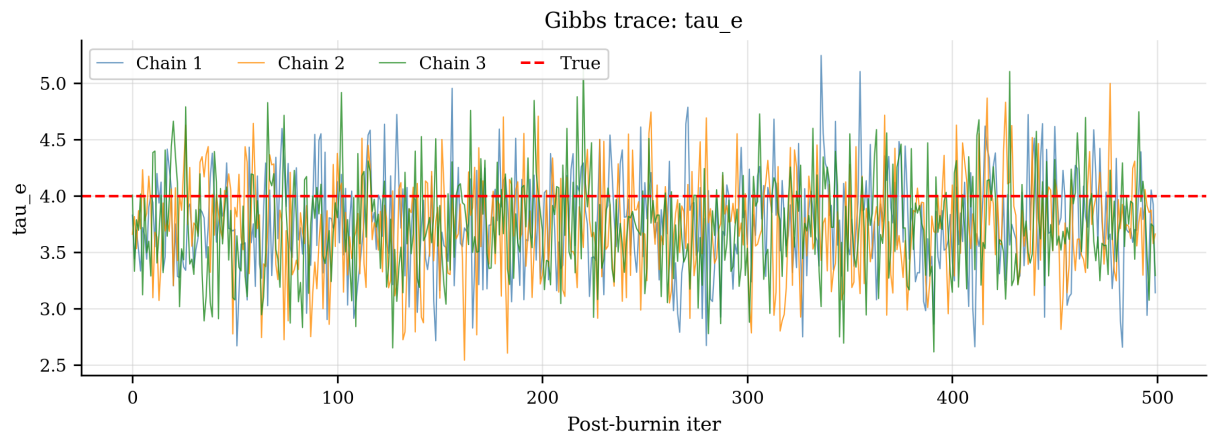


Figure 10: Gibbs trace: τ_e (noise precision).

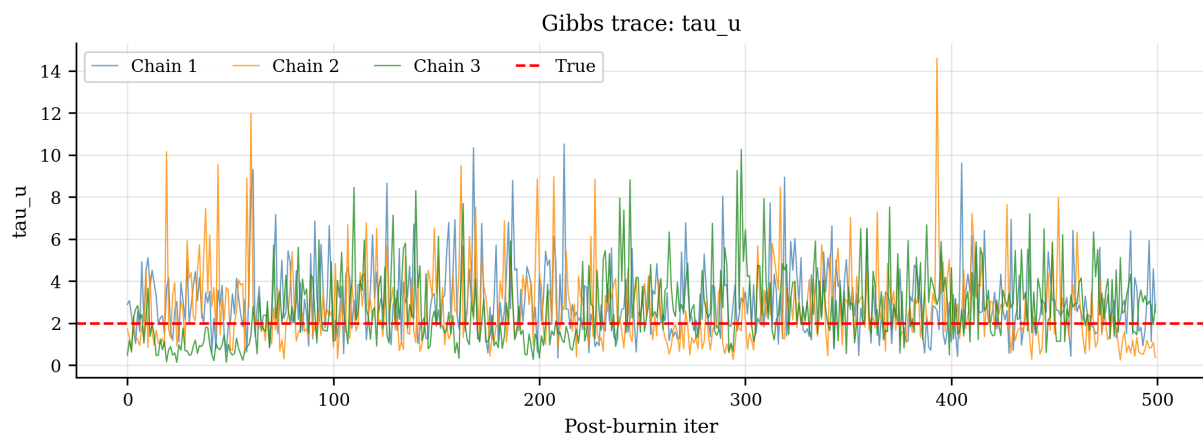


Figure 11: Gibbs trace: τ_u (random-effect precision).

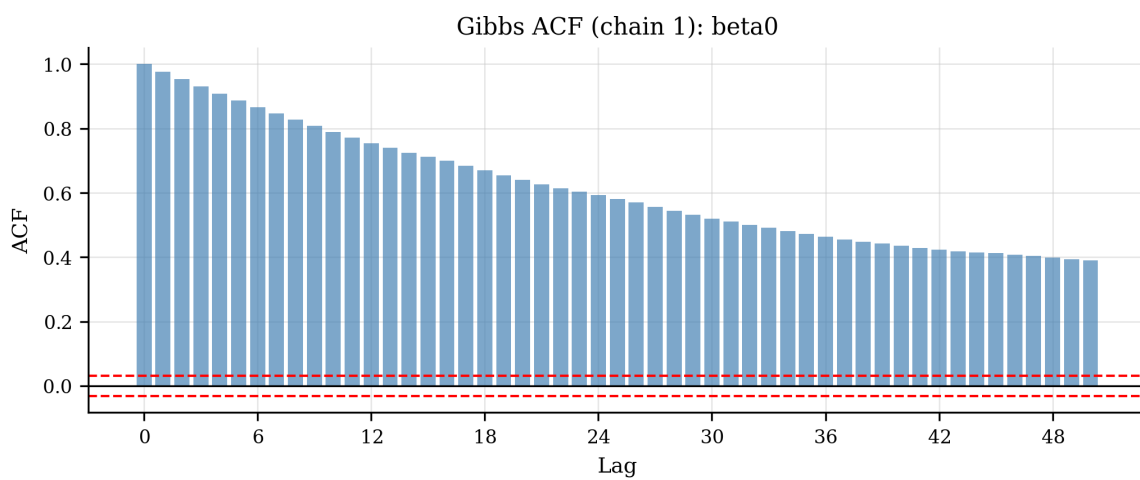


Figure 12: ACF for β_0 (chain 1). Red dashed lines: 95% band.

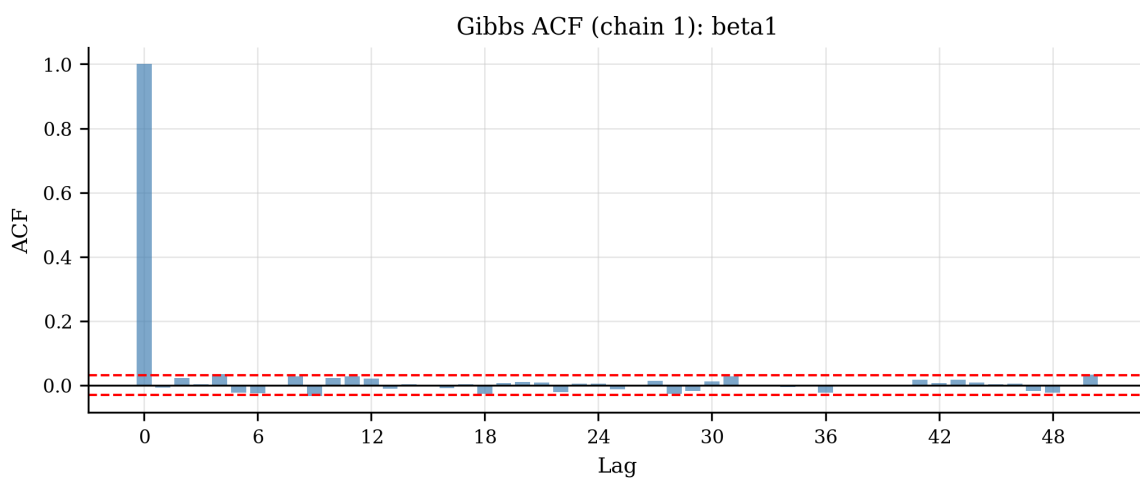


Figure 13: ACF for β_1 .

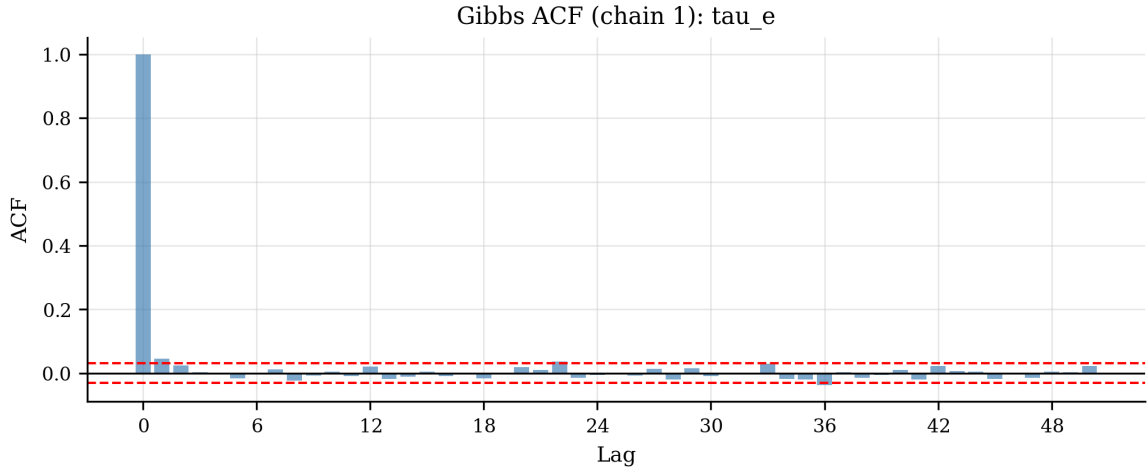


Figure 14: ACF for τ_e .

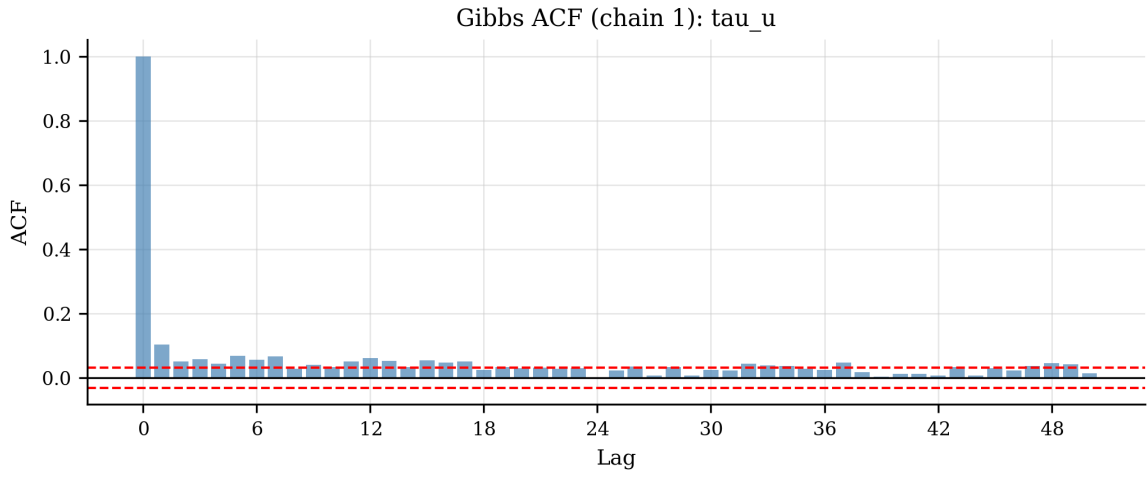


Figure 15: ACF for τ_u .

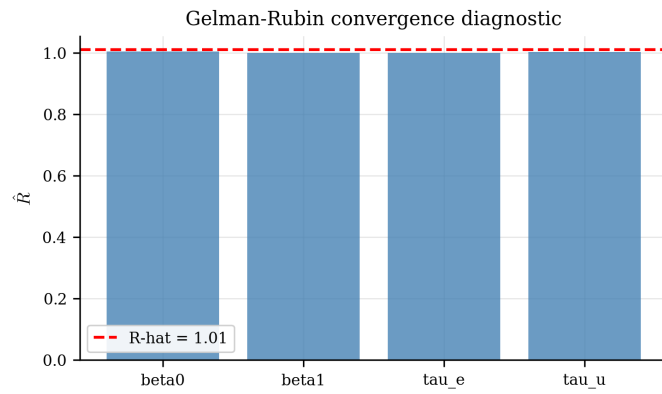


Figure 16: Gelman–Rubin $\hat{R}$ diagnostic for all monitored parameters. All values below 1.01 confirm convergence.

7.5 Random Effects Recovery

Figure 17 compares the CAVI posterior means for u_j against the Gibbs reference and the true values. Both methods recover the group-specific shifts, with CAVI showing slightly tighter

credible intervals consistent with its general tendency towards variance collapse.

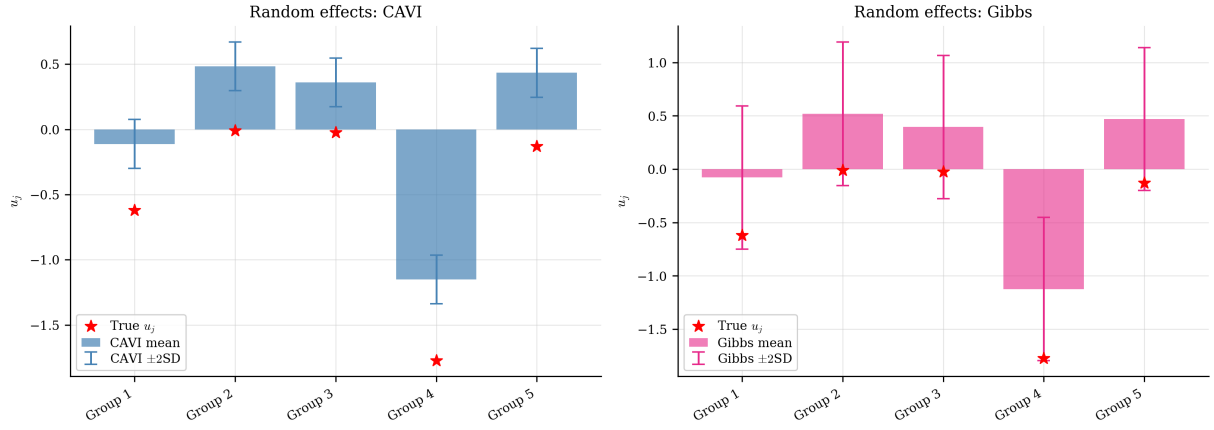


Figure 17: Random effect estimates u_j for CAVI (left) and Gibbs (right). Stars indicate true values; error bars show ± 2 posterior SD.

7.6 Posterior Comparison: VI versus Gibbs

Figures 18–21 overlay the VI approximations against the Gibbs histograms for the four scalar parameters.

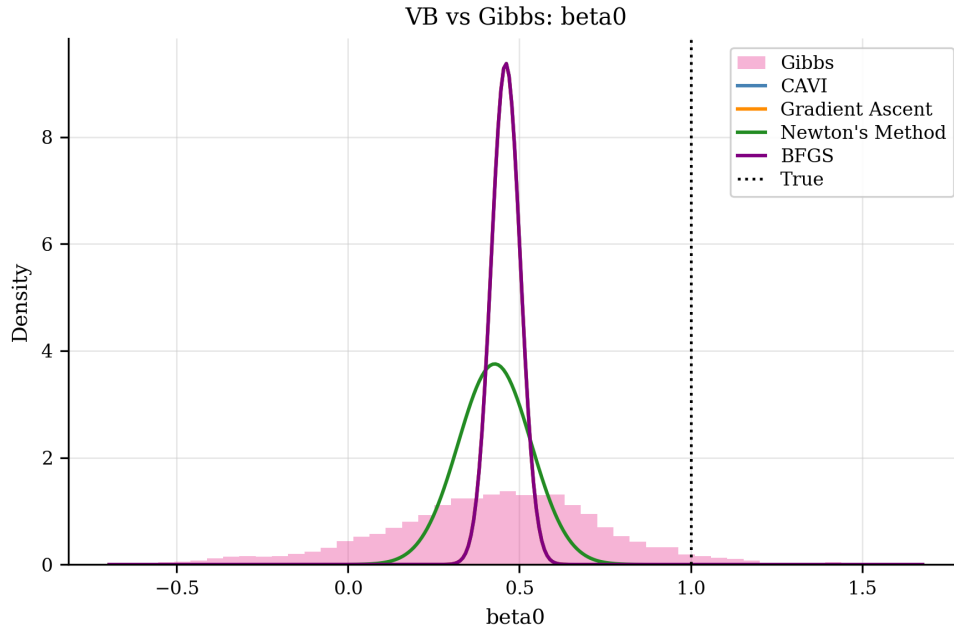


Figure 18: Posterior comparison for β_0 : Gibbs histogram versus VI Gaussian approximations. The VI curves are dramatically narrower than Gibbs (collapse).

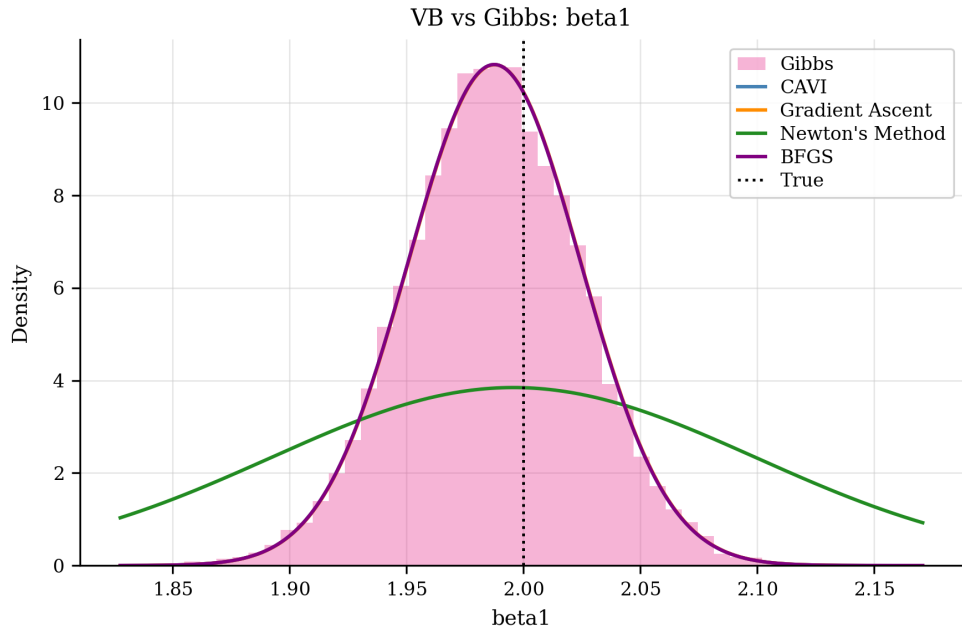


Figure 19: Posterior comparison for β_1 . VI and Gibbs agree well; minimal collapse.

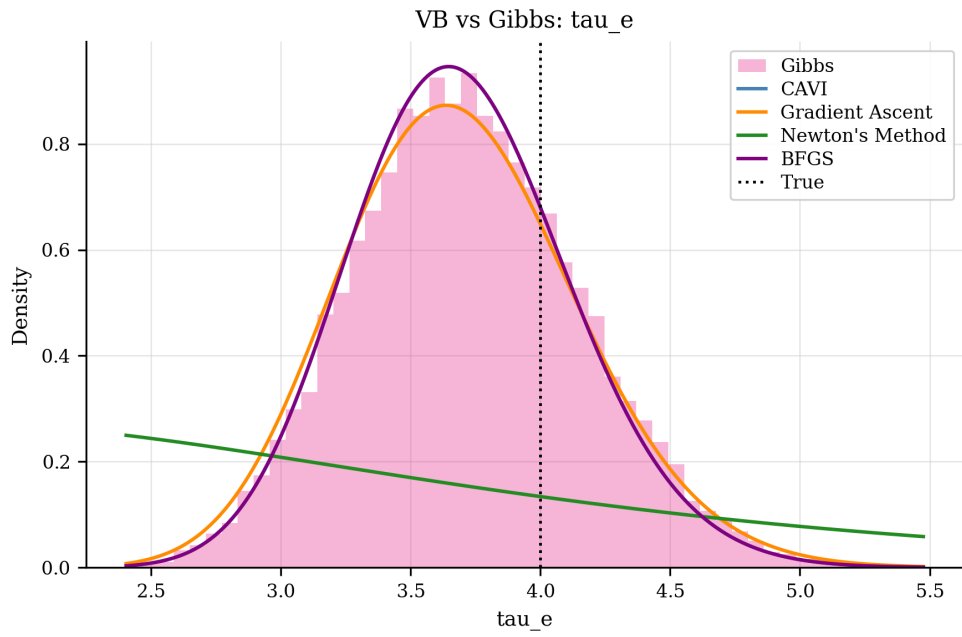


Figure 20: Posterior comparison for τ_e .

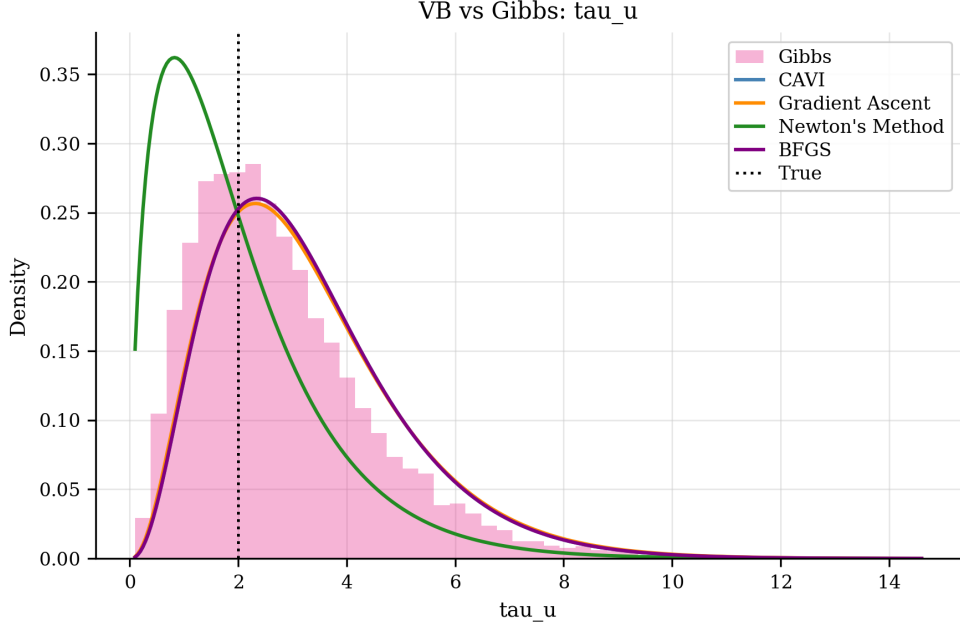


Figure 21: Posterior comparison for τ_u .

7.7 Posterior Standard-Deviation Ratios

Figure 22 shows the ratio $\sigma_{\text{VI}}/\sigma_{\text{Gibbs}}$ for each method and parameter. CAVI and BFGS exhibit severe collapse for β_0 (SD ratio ≈ 0.13) while maintaining near-unity ratios for β_1 . This asymmetric collapse arises because β_0 (the intercept) is confounded with the group means u_j under the mean-field factorisation: the posterior covariance between β_0 and $\mathbf{u}$ is discarded, causing $q(\beta_0)$ to appear far more concentrated than the true marginal. Newton's method produces anomalous SD ratios ($\beta_1 \approx 2.8$) due to its convergence to a suboptimal point.

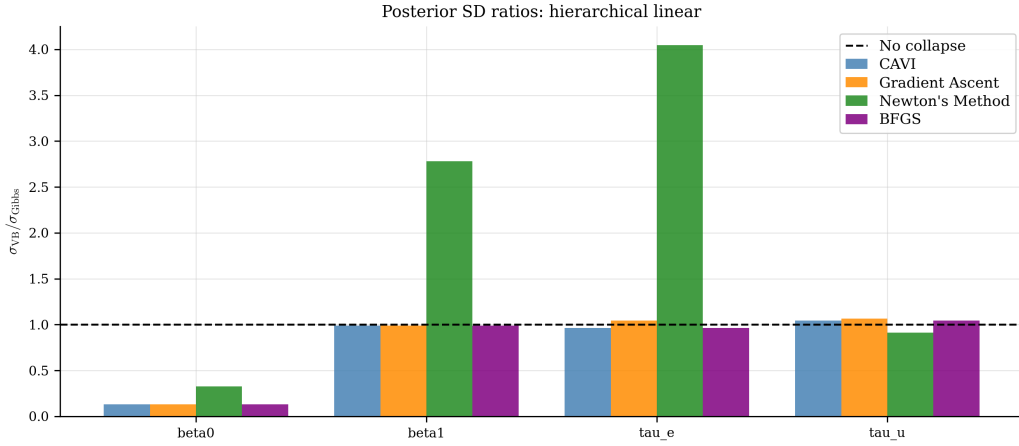


Figure 22: Posterior SD ratios (VI / Gibbs). Dashed line: ideal ratio 1.0. Severe intercept collapse (≈ 0.13) for CAVI, gradient ascent, and BFGS.

7.8 Posterior Mean Errors

Figure 23 shows the absolute error $|\mu_{\text{VI}} - \mu_{\text{Gibbs}}|$. Despite severe variance collapse for β_0 , posterior means are well recovered by all converged methods.

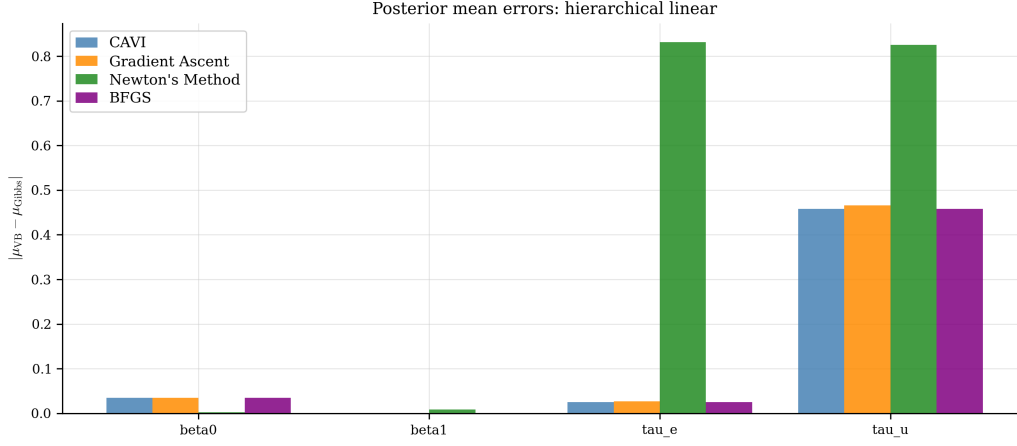


Figure 23: Absolute posterior mean errors relative to the Gibbs reference. Posterior means are accurate even where variance is collapsed.

7.9 Performance Summary

Table 1 summarises the convergence, runtime, and final ELBO for all methods. Table 2 summarises the posterior means and SD ratios.

Table 1: Optimisation performance: hierarchical linear regression.

Method	Iterations	Runtime (ms)	Speedup vs Gibbs	Final ELBO
CAVI	102	26.8	134x	-132.077
Gradient Ascent	2000	8890.5	0x	-132.086
Newton's Method	50	3877.6	0.9x	-165.282
BFGS	34	116.9	31x	-132.077
Gibbs (3 chains)	3x5000	3587.7	1x (reference)	N/A

Table 2: Posterior summary: hierarchical linear regression.

Method	μ_{β_0}	μ_{β_1}	$E[\tau_e]$	$E[\tau_u]$	SD ratio β_0	SD ratio β_1
CAVI	0.4604	1.9877	3.6959	3.2818	0.130	0.987
Gradient Ascent	0.4602	1.9878	3.6945	3.2895	0.130	0.988
Newton's Method	0.4280	1.9959	2.8896	1.9989	0.325	2.779
BFGS	0.4603	1.9877	3.6959	3.2818	0.130	0.987
Gibbs (ref)	0.4257	1.9872	3.7214	2.8239	1.000	1.000

7.10 Computational Cost

Figure 24 compares the runtimes of all methods. CAVI is the fastest converged method at 27 ms, $134\times$ faster than the conjugate Gibbs sampler (3.6 s for 3 chains $\times$ 5000 iterations). BFGS is $31\times$ faster than Gibbs. Gradient ascent reaches the same final ELBO as CAVI but requires the full 2000-iteration budget, making it $\sim 3\times$ slower than Gibbs at this problem size. Newton's method, despite only 50 iterations, requires a 19×19 FD Hessian at each step (requiring $19^2 = 361$ ELBO evaluations per iteration), giving a total cost comparable to Gibbs.

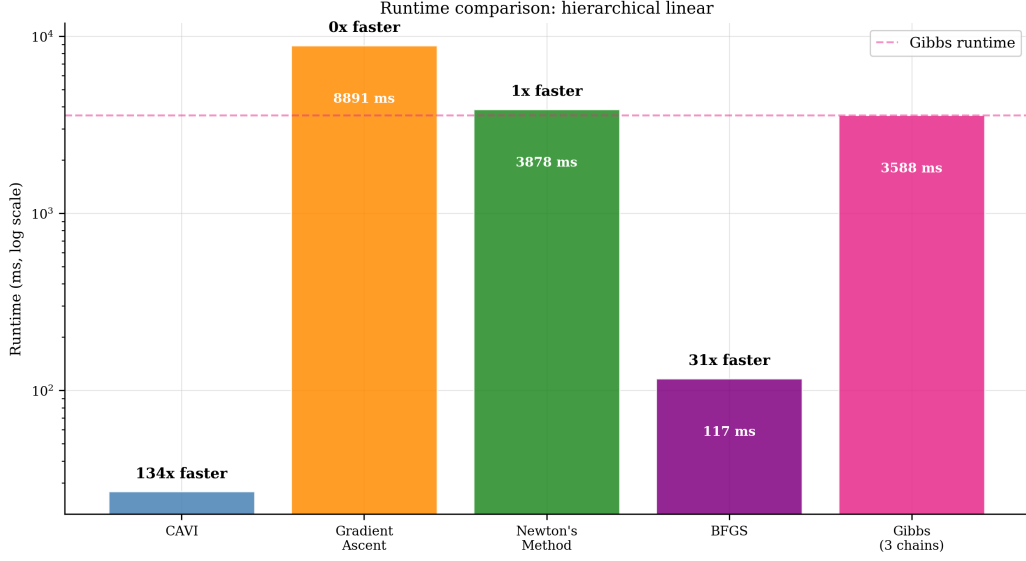


Figure 24: Runtime comparison (log scale). Annotated speedup factors relative to Gibbs.

8 Discussion

Posterior variance collapse. The intercept β_0 exhibits severe collapse (SD ratio ≈ 0.13) under all well-converged VI methods. The mechanism is the mean-field assumption: the true posterior has strong negative correlation between β_0 and the group means $\bar{u} = J^{-1} \sum_j u_j$, because the data constrain $\beta_0 + \bar{u}$ far more tightly than either component alone. Mean-field VI factorises these two blocks, forcing $q(\beta_0)$ to be a narrow Gaussian centred at the marginal mode rather than the true marginal. The slope β_1 is not confounded with the random effects in the same way, so its VI standard deviation matches Gibbs closely.

Newton's method failure. Newton's method converges to a point with ELBO = -165.28 , substantially worse than the global CAVI optimum at -132.08 . This is a consequence of the finite-difference 19×19 Hessian having poor conditioning throughout the trajectory: the Hessian of the ELBO mixes parameters that operate on very different scales (μ_β in $[-2, 2]$, $\log b_u$ near $[-3, 3]$), making the Newton step unreliable even with regularisation. BFGS avoids this issue by building an *adaptive* positive-definite inverse-Hessian approximation, which effectively rescales each parameter direction from curvature information accumulated over iterations.

CAVI versus BFGS. CAVI and BFGS converge to the same ELBO optimum with comparable accuracy. CAVI is $\sim 4\times$ faster in this implementation because each iteration requires only closed-form matrix operations, whereas BFGS requires $2D = 38$ ELBO evaluations per iteration for FD gradients. For models where CAVI updates are not available (e.g. non-conjugate likelihoods), BFGS is the natural first-choice gradient-based method.

Gradient ascent. First-order gradient ascent converges to the correct optimum but requires all 2000 iterations, making it the slowest converged method. The Armijo step sizes (Figure 6) show consistently small values, indicating that the curvature of the ELBO surface is poorly aligned with the gradient direction in a 19-dimensional space — the same limitation identified in the simpler linear and quadratic case studies, now more pronounced.

9 Conclusion

Hierarchical Bayesian linear regression with $J = 5$ groups and $D = 19$ unconstrained parameters presents a qualitatively more challenging optimisation problem than the two-parameter logistic case. CAVI and BFGS both find the global mean-field ELBO optimum and recover accurate posterior means, but both exhibit severe variance collapse for the intercept β_0 due to the mean-field factorisation ignoring the posterior correlation between the intercept and the random effects. Newton’s method fails to converge to this optimum, highlighting the limitations of finite-difference second-order methods when the Hessian is ill-conditioned. Gradient ascent converges but is an order of magnitude slower. The conjugate Gibbs sampler provides accurate marginal posteriors but at a cost $> 100\times$ that of CAVI for this problem size.